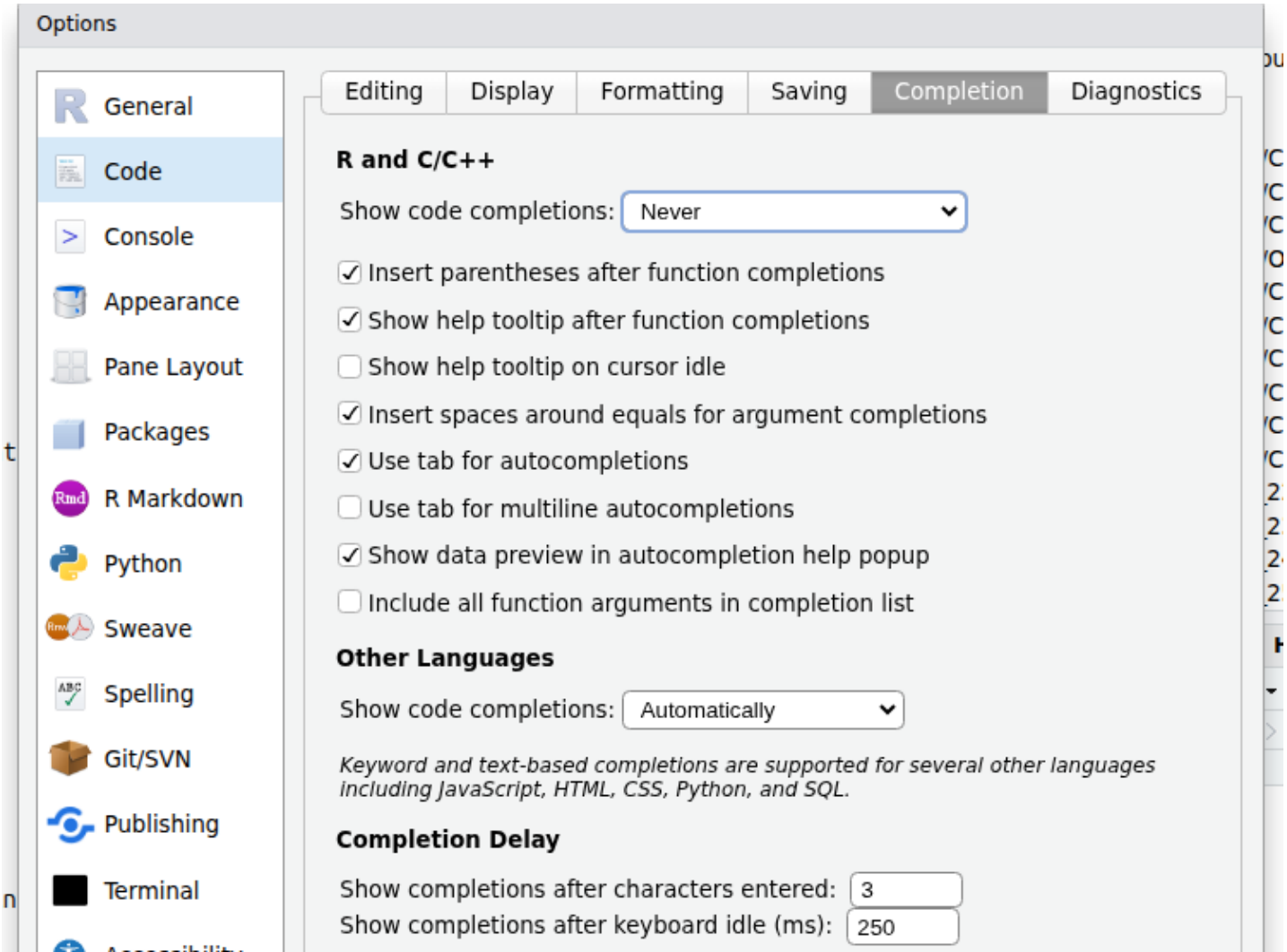
Lab 9: Search and Iterative algorithms

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Rstudio set up (ignore if you have already done this)

Rstudio has auto-suggestion features which we want to disable.

To do that go to `Tools` > `Global Options` > `Code` > `Completion` and set `Show code completions` to `Never`.



Homework 1

Starting from the code provided during the lectures:

1. Complete the code for the hill-climbing algorithm.
2. Test it on the objective function

```
f <- function(x) {
  return( sin(x/2) + cos(x/4) )
}
```

3. Experiment with other values of `epsilon` and with other objective functions.

4. Recall that the function `seq` works as follows:

```
seq(0, 1, by = 0.1)
```

```
 [1] 0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0
```

```
seq(-1, 1, by = 0.5)
```

```
[1] -1.0 -0.5  0.0  0.5  1.0
```

and look at what the `which.max` function does:

```
which.max( c(1.1, 2.3, 0.1) )
```

```
[1] 2
```

```
which.max( c(2.3, 0.1, 1.1) )
```

```
[1] 1
```

Using these functions, write a vectorised version of `subproblem` that does not include any loop.

Homework 2

Start from `lab_ttt_template.R`.

1. Complete the functions needed to play manually:
    - `create_board`
    - `print_board`
    - `play_move` (already done)
    - `get_elem`
    - `is_playable`
    - `set_elem`
2. Test whether these functions work.

Homework 2

3. Complete the functions needed to play automatically:
   - `ai_play`
   - `evaluate_move` (already done)
   - `pieces_at_row`
   - `pieces_at_col`
   - `pieces_at_diag`
   - `pieces_at_anti_diag`
4. Test whether it works by initialising from, e.g.:

```
X O *
* * *
* * *
```

and then play `X` and `O` in turn, until one wins.

Homework 3

Here we revisit hill-climbing.

1. Open `lab_steep_template.R`
2. Implement the hill-climbing algorithm according to the skeleton code.
3. Verify your implementation with the example output provided in the code comments.
4. By changing `x0`, could you get hill-climbing stuck on one of the local optima?
    - By changing `epsilon` (and keeping `x0`), could the algorithm recover the global optimal solution again?

Homework 4

1. Implement the steepest ascent algorithm according to the skeleton code.
2. Verify your implementation with the example output provided in the code comments.
3. Using the same `x_0` and `epsilon`, do the two algorithms produce the exact same solution?
    - If not, how different are they?
    - Which one produces a better solution?
4. Change the initial guess, observe gradient ascent produces a locally optimal solution.
5. Check how gradient ascent works for very large and very small `epsilon`.

Submission

Submit your completed R solutions files for Homework 2 and 3.

Before submitting, make sure your code runs.